

ABD600 Project

University West 2026

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Project

This project extends the work from the Labs, focusing on Automated Guided Vehicles (AGVs) within a factory setting.

The core objective is to demonstrate the integration of PLC programming with Big Data and IoT by navigating an AGV through a predefined factory route.

Grading

See course PM and course plan for complete course grading.

This project is graded in the following levels:

Fail

If not Pass or higher

Pass

Requirement R1, R2, R3, R4 implemented

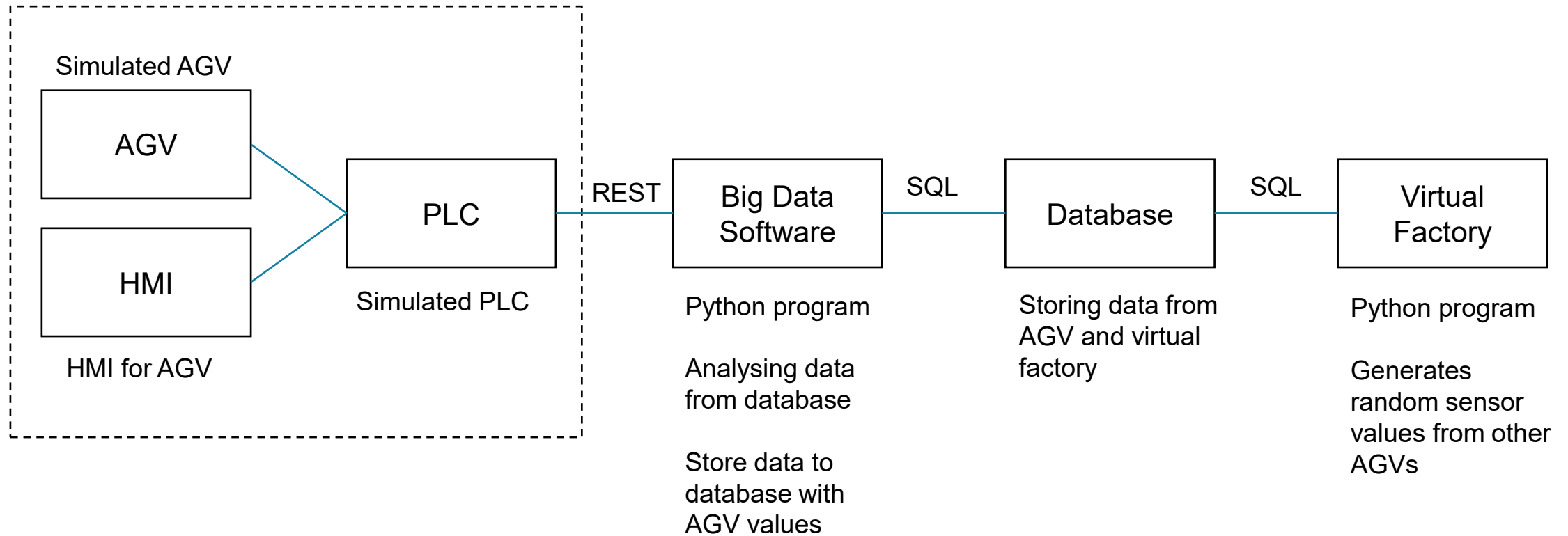
Examination

You need to follow the project requirements.

You show your results by running your software live and explaining it to a teacher on the course.

Questions will be asked, and you must be able to answer them, showing that you understand your project solution.

Overview



R1: AGV Movement

- ❑ Program the AGV to go to these checkpoints:

Home > A > B > C > D > Home.

R2: Capturing data

- ❑ At each location, the AGV must pause and transmit its time stamp, battery and location (A, B, C, D, Home) to the database.

R3: Data analysis

- ✓ The AGV moves some items that are sensitive to temperature and humidity.
 - ✓ The script *VirtualFactory.py* generates random temperature and humidity values every second, and stores them in the database, mimicking readings from various factory locations from other AGVs.
 - ✓ Each data point includes a "damage/no damage" indicator, reflecting the condition of temperature-sensitive items transported by the AGV.
 - ✓ A location is deemed safe if no damage incidents have been recorded in its 10 most recent measurements.
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- ❑ Implement logic to assess the safety of each location based on the latest 10 data points in the database.

R4: AGV Safety

- ❑ If a location is marked unsafe, the AGV must wait until it is safe to proceed.
- ❑ Implement clear states describing the AGV safety state.

Visualization – Not graded

- ❑ The PLC's Human Machine Interface (HMI) displays the safety status for each location (A, B, C, D).
- This visualization exercise is not mandatory!



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